

Experiment 1: Advanced 8-bit ALU with NZCV flags

1. Objective

To design, simulate, and implement an Advanced 8-bit Arithmetic Logic Unit (ALU) capable of performing various arithmetic and logical operations while generating Negative (N), Zero (Z), Carry (C), and Overflow (V) flags.

2. Block Diagram

..



Syntax error in text
mermaid version 12.1.0

Parse error on line 6: ...gs[NZCV Flags 4-bit]-p-</p><pre><code>s
.....^ Expecting 'SEMI', 'NEWLINE', 'SPACE', 'EOF',
'SHAPE_DATA', 'STYLE_SEPARATOR', 'START_LINK', 'LINK', 'LINK_ID', got
TAGSTART